Bayes networks

Tools of probabilistic reasoning

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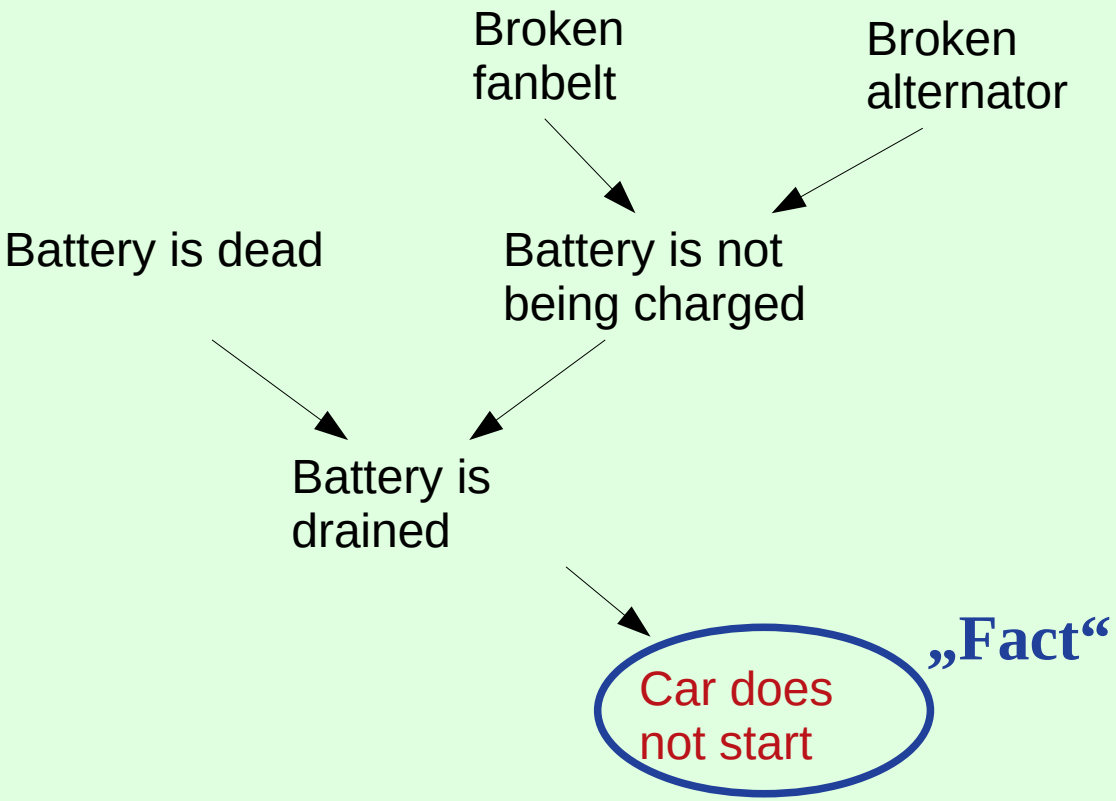
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Partially observable environment

- Single observation does not uniquely determine state
  - *Multiple observations* are needed
  - *Environment model* is required to store information

- Today we'll talk about *probabilistic causal* models
  - State is described as a collection of *events* (statements)
  - Events can be linked by *causal relations*
  - Relation strength is measured by a *probability P* ~ [0;1]

The cold light of day

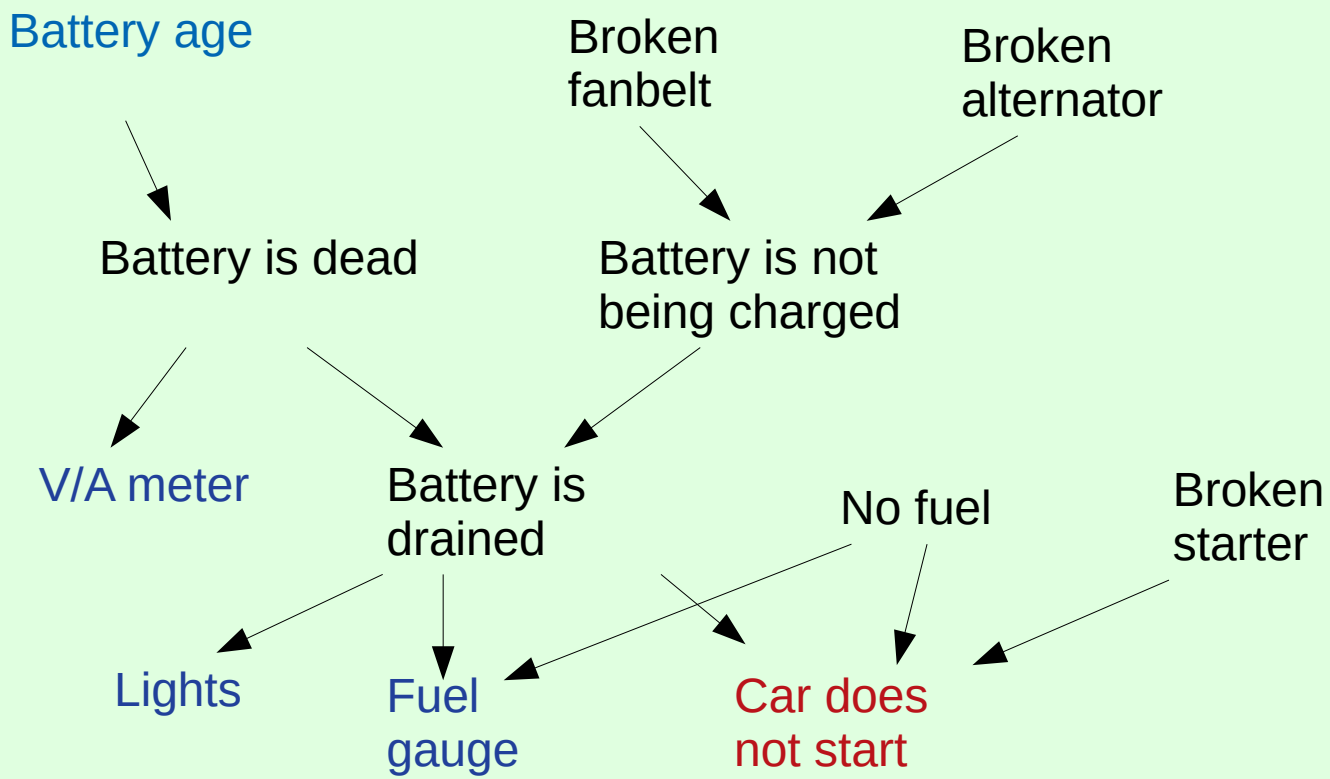


Possible causes

„Fact“

Causes and effects

Observable events



Bayes network

- A representation of probability distribution
- For now, we will only consider binary events: $X = \{T,N\}$
  - What is the *probability* of state from N events?
  - How many *variables* describe such a state?

- Causal relations reduce the number of needed variables
  - Internal structure is taken into account
  - Can you guess the number of independent variables required to fully describe a state of the network in slide 4?

Probability cheat sheet

Sum of probabilities:

$$\sum_i P(x_i) = 1$$

Joint probability:

$$P(x, y, \ldots)$$

Independent variables:

$$P(x, y) = P(x)P(y)$$

Conditional probability $P(y|x)$:

$$P(x, y) = P(y|x)P(x)$$

Sum over conditions:

$$P(y) = \sum_i P(y|x_i)P(x_i)$$

Bayes theorem:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Conditionally independent vars x and y given z:

$$P(x, y|z) = P(x|z)P(y|z) \quad \text{or} \quad P(x|y, z) = P(x|z)$$

Sum over conditions

- Events X_1 and X_2 are defined as follows:
  - $P(X_1=T) = 0{,}5$
  - $P(X_2=T \mid X_1=T) = 0{,}9$, $P(X_2=T \mid X_1=N) = 0{,}2$
- What is the value of $P(X_2=T)$?

  - $P(X_2=T) =$

    $= P(X_2=T|X_1=T) \cdot P(X_1=T) + P(X_2=T|X_1=N) \cdot P(X_1=N) =$

    $= 0{,}9 \cdot 0{,}5 + 0{,}2 \cdot 0{,}5 = 0{,}55$

Joint probability

- Test results (to determine K) are defined like this:
  - $P(t, K) = 0{,}009$
  - $P(n, K) = 0{,}001$ $[n \equiv \neg t]$
  - $P(t, \neg K) = 0{,}198$
- What is the value of $P(K \mid t)$?

$$P(K|t) = \frac{P(t,K)}{P(t)} = \frac{P(t,K)}{P(t,K)+P(t,\neg K)} = \frac{0{,}009}{0{,}009+0{,}198} \approx 0{,}043$$

Conditional probability

- Test results (to determine K) are *now* defined like this:
  - $P(K) = 0{,}01$
  - $P(t \mid K) = 0{,}9$
  - $P(n \mid \neg K) = 0{,}8$
- Again, what is the value of $P(K \mid t)$?

$$P(K|t) = \frac{P(t|K)\cdot P(K)}{P(t)} = \frac{P(t|K)\cdot P(K)}{P(t|K)\cdot P(K)+P(t|\neg K)\cdot P(\neg K)} =$$

$$= \frac{0{,}9\cdot 0{,}01}{0{,}9\cdot 0{,}01+0{,}2\cdot 0{,}99} = \frac{0{,}009}{0{,}009+0{,}198} \approx 0{,}043$$

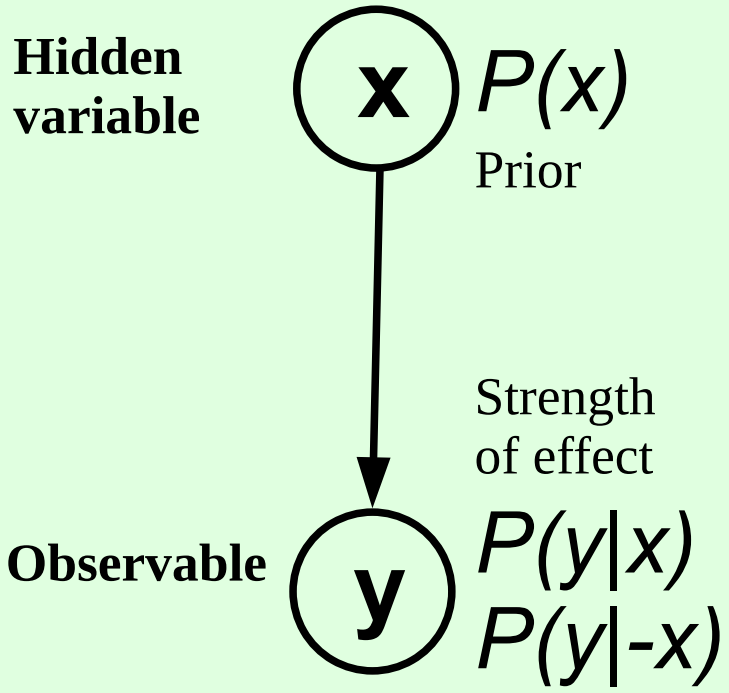
Bayes network explained

- Bayes theorem (rule):

$$P(x|y)=\frac{P(y|x)P(x)}{P(y)}$$

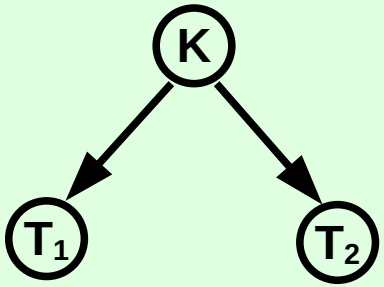
  - y – evidence (effect)
  - x – event (cause)

  - $P(y|x)$ – **causal** relation (known information)
  - $P(x|y)$ – **diagnostic** relation (new information!)



Double test case

- The test to determine K is performed twice
  - $P(K) = 0{,}01$
  - $P(t \mid K) = 0{,}9$, $P(n \mid K) = 0{,}1$
  - $P(n \mid \neg K) = 0{,}8$, $P(t \mid \neg K) = 0{,}2$
- What is the value of $P(K \mid t,t)$?



$$P(K|t_1 t_2) = \frac{P(K)\cdot P(t_1|K)\cdot P(t_2|K)}{P(t_1 t_2)} =$$

$$= \frac{0{,}01\cdot 0{,}9\cdot 0{,}9}{0{,}01\cdot 0{,}9\cdot 0{,}9 + 0{,}99\cdot 0{,}2\cdot 0{,}2} \approx 0{,}1698$$

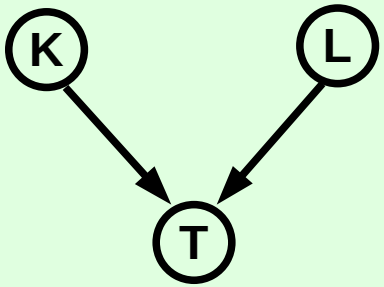
- …but only if a certain assumption holds!

Conditional independence

- Postulated by the following expressions:
  - $P(t_2 \mid K,t_1) = P(t_2 \mid K)$
  - $P(t_1,t_2 \mid K) = P(t_1 \mid K) \cdot P(t_2 \mid K)$
- Such events are independent *given* value of condition K
  - If K is known, t_1 result does not influence t_2
- Are t_1 and t_2 „ordinarily“ independent?

Multiple-cause event

- Consider Bayes network with parameters below

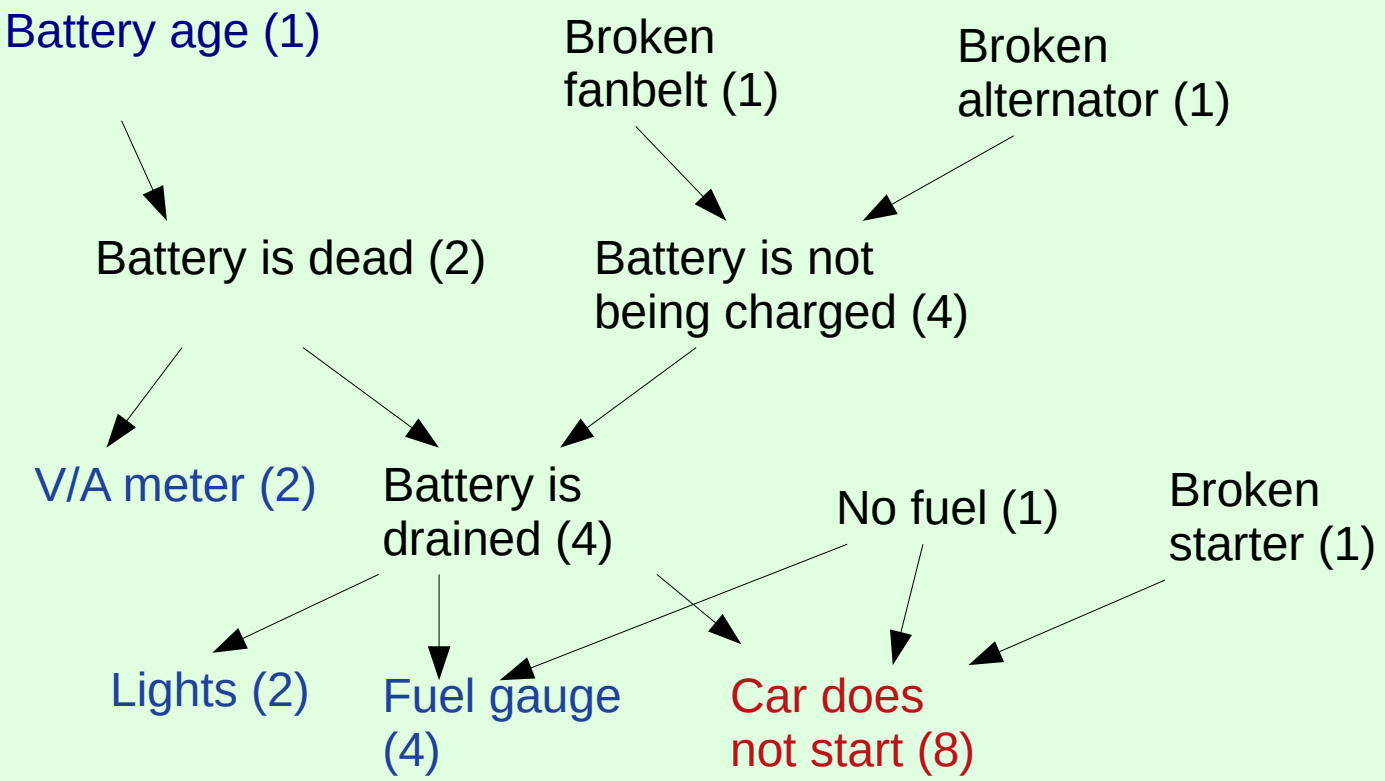


- a) Are L and K independent?
- b) Are L and K *conditionally* independent?

  - $P(K) = 0{,}7$
  - $P(L) = 0{,}01$
  - $P(T|K, L) = 1$
  - $P(T|\neg K, L) = 0{,}9$
  - $P(T|K, \neg L) = 0{,}7$
  - $P(T|\neg K, \neg L) = 0{,}1$

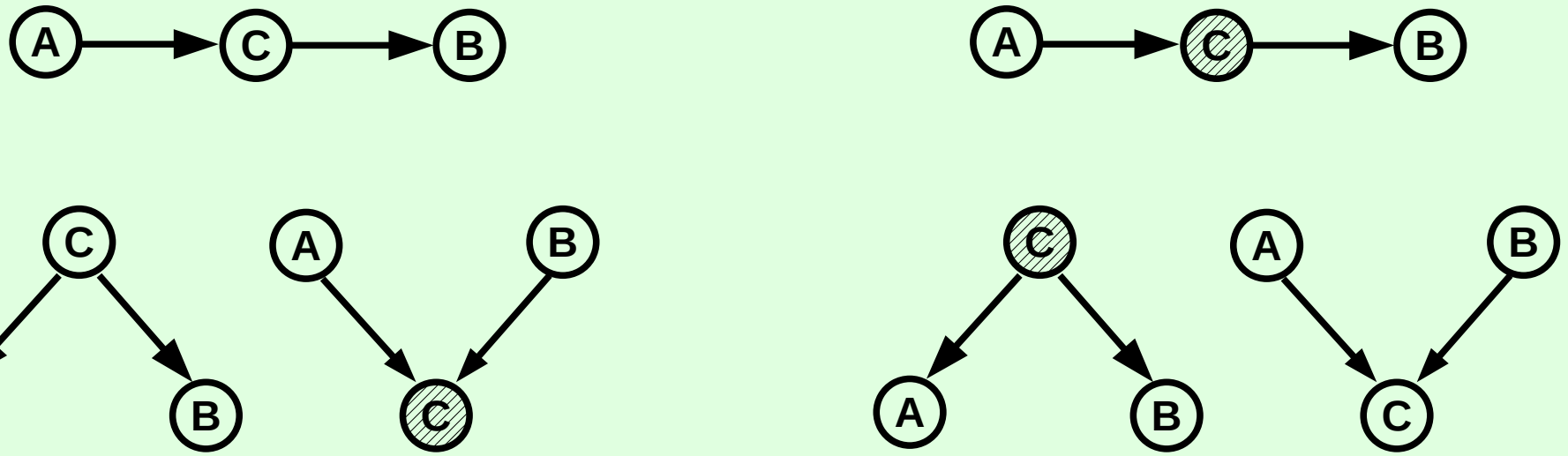
Answer: variables of certain Bayes network

Total:
31 variable



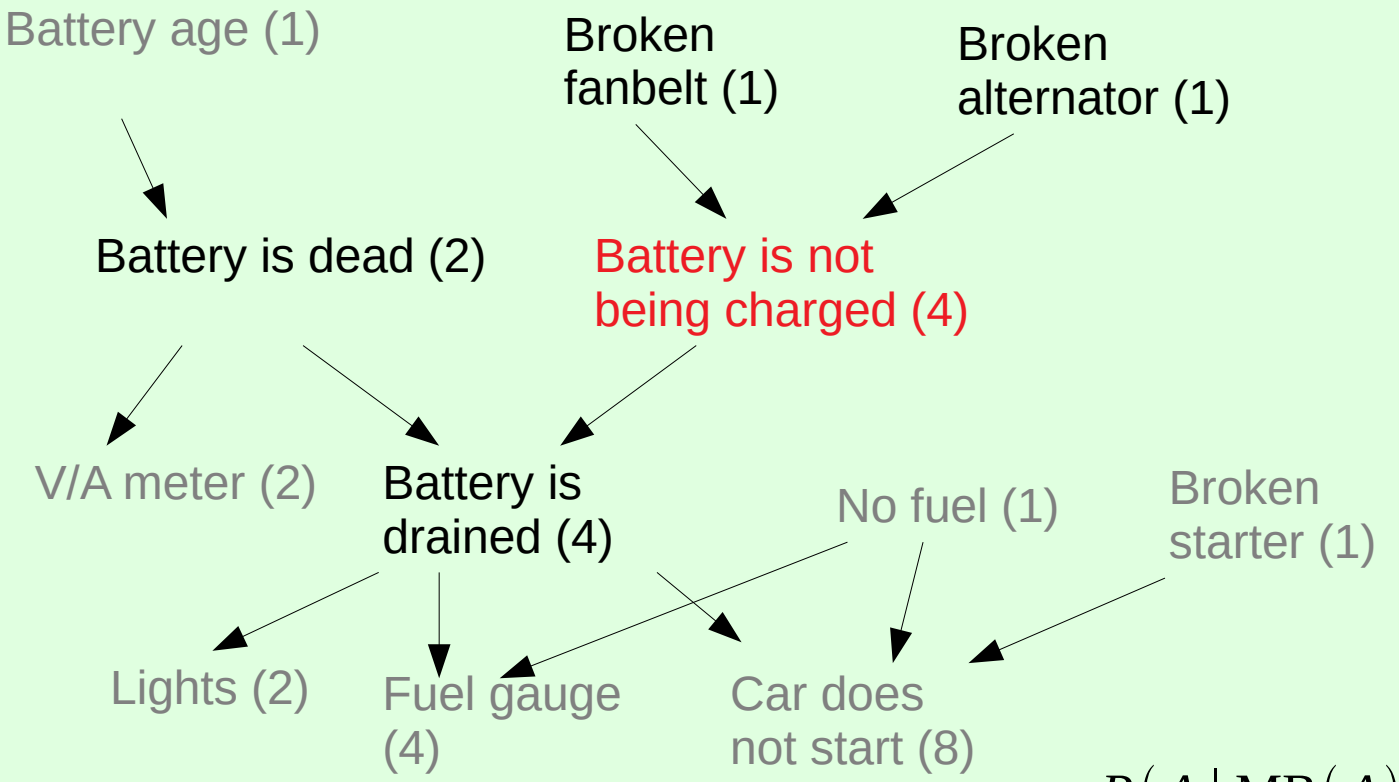
d-separation

- A method to split Bayes network into sections
  - Less dependent elements → less variables to describe them
- Graphical depiction of variable dependence
  - On the left A and B – *dependent*, on the right – *independent*
  - Shaded elements correspond to known values



Markov blanket

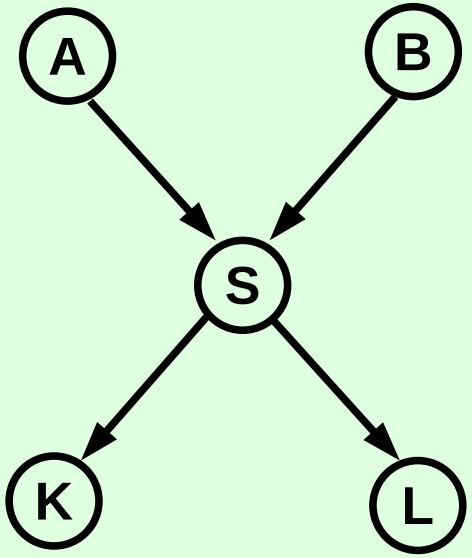
- *d*-separation rules suggest that every node in the network is only (conditionally) dependent to its parents, children and (...spouses?)
- These nodes define *Markov blanket*



$$P(A \mid \mathrm{MB}(A), B) \equiv P(A \mid \mathrm{MB}(A))$$

Inference in Bayes networks

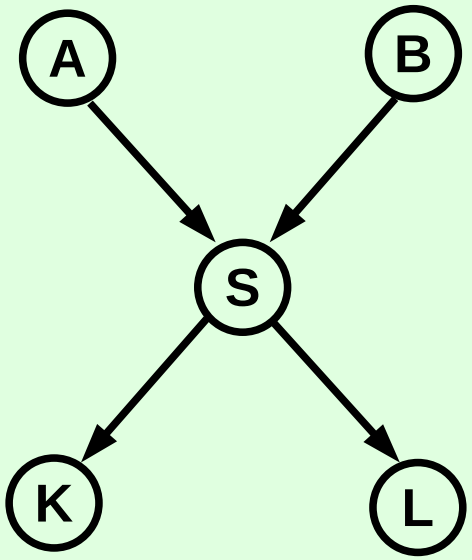
- All nodes in network are one of:
  - Evidence (E)
  - Query (Q)
  - Hidden (H)
- Desirable output:
  - Distribution $P(Q_1,Q_2...|e_1,e_2...)$
  - Q values, where $P(Q|e)=P_{\max}$
- How to compute Q values?



Enumeration

- Consider $P(+a|+k,+l)$:
  - $P(+a|+k,+l) = P(+a,+k,+l)/P(+k,+l)$;
  - $P(+a,+k,+l) = \sum_b \sum_s P(+a,+k,+l,b,s) =$

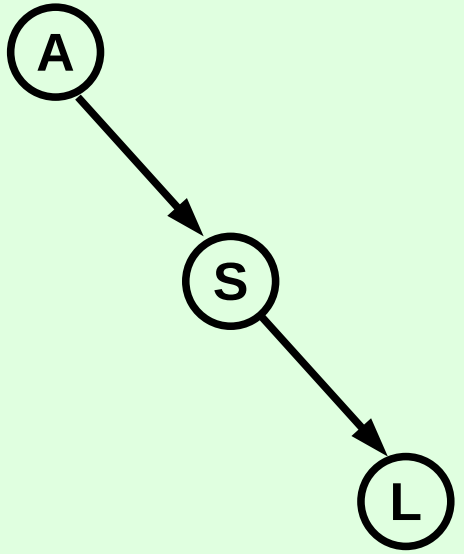
    $= \sum_b \sum_s P(+a)\cdot P(b)\cdot P(s|+a,b)\cdot P(+k|s)\cdot P(+l|s)$



- Final probability – sum of products
- Exact, but not always practical
  - When the network is very large
  - When the relational structure is poor

Variable elimination

- Consider *P(+l)*
  - $P(+l) = \sum_a \sum_s P(a) \cdot P(s|a) \cdot P(+l|s)$
- 1 step: joining factors
  - *P(a)P(s|a) = P(s,a)*
- 2 step: reduction
  - *P(s,a) → P(s)*
- Repeat until only desired variables remain
  - *P(s)P(+l|s) = P(+l,s) → P(+l)*



- Same as enumeration, but usually more efficient
  - Lower memory costs, slower error accumulation

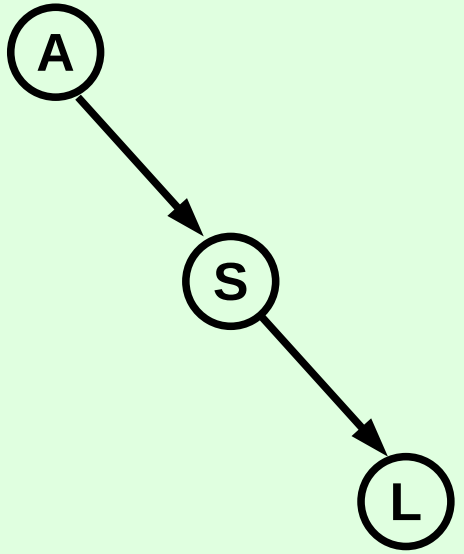
Approximate inference

- Probability values are determined by random sampling
  - Distribution is drawn from underlying Bayes network
  - $P(E_1=e_1, E_2=e_2...) = N(E_1=e_1, E_2=e_2...) / N_{total}$

- If known variables are correct, query values also converge
  - More efficient for large networks, also for „quick“ estimation

Random sampling

- Network variables:
  - $P(+a) = 0{,}8$
  - $P(+s|+a) = 0{,}7$, $P(+s|\neg a) = 0{,}6$
  - $P(+l|+s) = 0{,}5$, $P(+l|\neg s) = 0{,}4$
- What is the value of $P(+l)$?
- Select values from uniform distribution
  - 0,47 0,61 0,74; 0,62 0,96 0,37; …
- Create joint samples
  - $+a$, if *random()* < 0,8, $\neg a$ otherwise
  - $(+a,+s,\neg l)$; $(+a,\neg s,+l)$; …
- Given first two samples, $P(+l) = 0{,}5$
- What is the exact value?



Rejection sampling

- A way to determine conditional probabilities
  - Conditional variables are kept fixed
- Consider $P(+l|+a)$

- Samples that do not match conditions are rejected
  - $(+a,+s,+l)$; $(+a,\neg s,+l)$; ~~$(\neg a,\neg s,\neg l)$~~; …

- Main drawback:
  - If desired conditions are rare, most samples are rejected and do not contribute towards estimation

Likelihood weighting

- Consider $P(+l|+a)$ again
  - Conditional variables are kept fixed

- All samples are accepted
  - But weighted according to the probability of conditions
  - $(+a,+s,+l) \cdot 0{,}8$; $(+a,\neg s,+l) \cdot 0{,}8$; $(\neg a,\neg s,\neg l) \cdot 0{,}2$; …

- Main drawback:
  - Slow progress while computing „upstream“ variables (fixed effects, queried causes)

Gibbs sampling

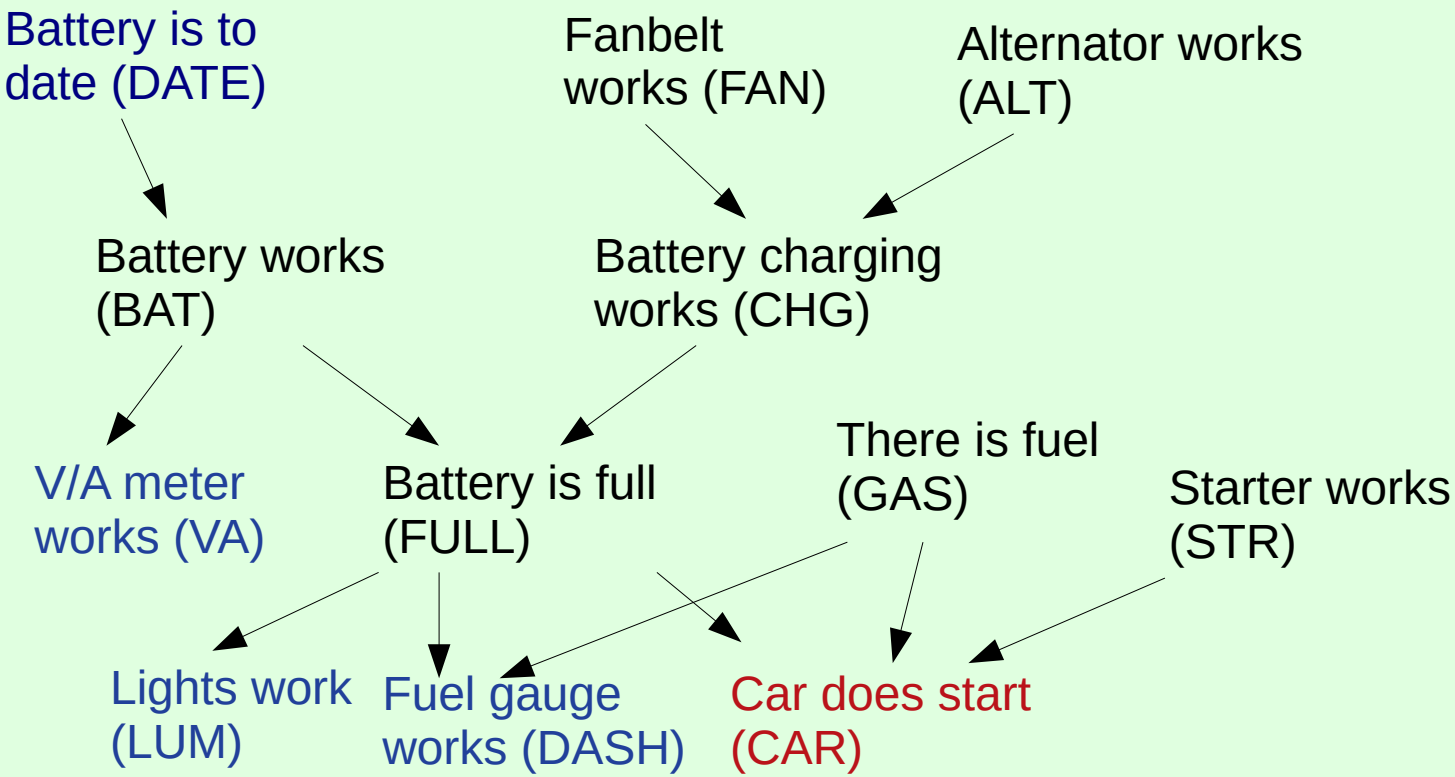
- Based on Monte Carlo Markov chains (*MCMC*)

- Consider $P(+l,+s|+a)$
  - Set initial sample (f. e., $(+a,\neg s,\neg l))$
  - Select a single query variable (e. g. s)
  - Regenerate query value given other sample values $((+a,\neg l) \rightarrow +s)$
  - Select another query variable and continue $((+a,+s) \rightarrow \neg l)$

- Similar to random walk in parameter space
  - In this case, variables are defined by their „Markov blanket“

3.2: clean-up of „car“ Bayes network

Total:
31 variable



3.2: inference in „car“ Bayes network

- Let the network parameters be the following:
  - DATE = 0,9; FAN = 0,98; ALT = 0,999; GAS = 0,95; STR = 0,995;
  - BAT|±DATE = 0,95/0,5; VA|±BAT = 0,99/0;
  - CHG|(FAN,ALT) = 0,99, *otherwise* 0;
  - FULL|(BAT,CHG) = 0,95, *otherwise* 0;
  - LUM|±FULL = 0,99/0;
  - DASH|(FULL,GAS) = 0,99, *otherwise* 0;
  - CAR|(FULL,GAS,STR) = 0,999, *otherwise* 0.

  - a) If the car does **not** start, how likely is the battery to be out of date?
  - b) Take the differently-connected network (next slide) and recompute its parameters from the given ones
  - c) Compare efficiency of a sampling algorithm of your choice to compute answer of a) by operating on the two Bayes networks

3.2: poorly (?) chosen Bayes network

Total:
35 variables

